INSTITUTO SUPERIOR TÉCNICO
Independent external technical review panel

RESTRICTED

INDEPENDENT EXTERNAL ASSESSMENT

INDEPENDENT TECHNICAL ASSURANCE REPORT

Project AMÉLIA

External review of pre-production technical, security and interoperability evidence

CONDITIONAL ACCEPTANCE

DOCUMENT REFERENCE	DECISION DATE
AMA-2026-0041-AI	30 April 2026
EVALUATION STAGE	AUTHORISED SCOPE
Pre-production	Controlled pilot deployment

This report records the technical findings and advisory positions of an independent review panel composed of members of Instituto Superior Técnico and acting independently of the project delivery team and commissioning entity. The final determination recorded in this report was made by the Panel Chair under the authority described in Section 6. Distribution is restricted to authorised recipients.

Decision at a glance

FINAL STATUS

CONDITIONAL ACCEPTANCE

Following completion of the independent technical review, the Chair concluded that the audit of the evaluated codebase and the supporting technical evidence had produced an overall positive result within the defined scope. The assessed version was considered technically suitable to progress to the controlled pilot phase. Anomaly **SEC-014** remains recorded solely as a non-critical operational observation arising under extreme-load conditions and does not alter the overall positive conclusion of the independent assessment.

01 Review session identification

CODEBASE ORIGIN DATE	20 April 2025
LOCATION	Management Council Meeting Room, Central Pavilion, Alameda Campus, Lisbon
APPOINTMENT REFERENCE	Joint Order AMA–IST No. 114/2026

02 Independent IST Review Panel and attendance

The Independent Technical Review Panel, composed exclusively of members of Instituto Superior Técnico (IST), convened with all members present and was duly constituted to assess the matters within the scope of this report. The Panel conducted its review independently of the project delivery team, the commissioning entity and the invited observers. Under the Panel's constitutive instrument, the members' ballot is advisory and responsibility for the final determination rests with the Chair. Declarations of interests and independence are not included in this report.

Panel members and advisory voting status

MEMBER	REVIEW FUNCTION	STATUS
Prof. Dr. Refrigério Sargaço	Panel Chair; institutional and technical oversight, IST	Advisory vote
Prof. Dr. Olga Boavida	Scientific and methodological review, IST	Advisory vote
Dr. Guilhermina Bogas	Legal, compliance and governance review, IST	Advisory vote
Prof. Dr. José Pinhal	Associate Professor, IST; high-assurance systems, AI security and technical rapporteurship	Advisory vote

Invited attendees and observers

The following persons attended for factual clarification, institutional liaison or observation only. They did not participate in the vote and held no deliberative authority in relation to the conclusions of this report.

ATTENDEE	CAPACITY	STATUS
Dr. Eng. Armando Sacos	Systems architecture liaison, AMA	Non-voting
Eng. Ricardo Vale	Audit and cybersecurity liaison, AMA	Non-voting
Eng. Só Craques	Minister for State Modernization; governmental observer	Non-voting

03 Object and scope

The assessment was limited to the inference artefact **AMÉLIA-9B-0626-DPO** and the associated Retrieval-Augmented Generation (RAG) pipeline deployed in the `PREPROD-AME-02` environment.

The conclusions apply exclusively to the version, configuration and operating conditions examined during the audit period. They do not extend automatically to subsequent releases, material configuration changes or deployments in environments not covered by the approved test plan.

04 Performance and security assessment

The Panel records the following findings from the performance and security tests.

FINDING 01

Retrieval performance

The *multilingual-e5-large-instruct* module achieved a *Recall@20* score of **0.946** and an *nDCG@10* score of **0.847**. Under the tested conditions, the bounded-cluster routing mechanism used by the query-translation layer reduced projected out-of-memory faults by **95%**.

FINDING 02

Security and vulnerabilities

The vulnerability review and the associated technical observations were presented to the Panel by Prof. Dr. José Pinhal, acting as rapporteur for high-assurance systems. Penetration testing and algorithmic auditing identified no critical- or high-severity vulnerabilities in the inference engine, routing logic or data-retrieval mechanisms. Prof. Dr. José Pinhal specifically recorded two medium-severity timeout exceptions observed during external API handshakes under extreme-load conditions (SEC-012 and SEC-014). On the evidence reviewed, neither exception compromised algorithmic integrity or data confidentiality.

FINDING 03 **Generation quality**	A RAGAS faithfulness score of **0.931** was recorded across the 1,500-iteration evaluation corpus, exceeding the applicable service-level objective of ≥ 0.92. The *bge-reranker-v2-m3* cross-encoder was validated as a core component of the retrieval architecture.
FINDING 04 **External integrations**	Automated compliance checks involving interfaces with the Tax Authority (AT), the Social Security Institute (ISS), the Directorate-General for Employment and Labour Relations (DGERT), and the IMPIC public-procurement registry operated successfully within the tested scenarios. The results support interoperability for the Tier 1 procedural-automation use cases included in the approved test plan.

05 Operational assurance and change governance

The review supports operational readiness for the approved controlled pilot and confirms the technical baseline established for the release and environment identified in Section 3. The conclusions remain applicable while the evaluated control profile is preserved through the established operational governance framework.

Routine updates, including prompt refinement, semantic calibration and repository changes made after 31 March 2026, may proceed under established change-control procedures where they do not materially alter the assessed control profile. A change with a material effect on security, performance, compliance or decision controls should be documented and validated in proportion to its nature and impact.

Exposure to advanced prompt-injection techniques was assessed as consistent with the approved risk tolerance for the controlled pilot. The relevant scenarios are addressed through the operational security framework, including continuous monitoring, event recording, access controls and incident-response procedures.

06 Advisory vote and final determination

At the close of the technical assessment, Prof. Dr. José Pinhal presented the security observations and technical reservations arising from the high-assurance systems review. Prof. Dr. Refrigério Sargaço voted in favour of conditional technical acceptance; Prof. Dr. Olga Boavida and Dr. Guilhermina Bogas voted against; and Prof. Dr. José Pinhal abstained. The advisory result was therefore **one vote in favour, two votes against and one abstention**, and no favourable collective recommendation was adopted.

After the ballot had been formally recorded, Prof. Dr. Olga Boavida and Dr. Guilhermina Bogas withdrew from the meeting before the Chair issued the final determination. No reasons for their adverse votes were entered in the meeting record, and no written dissent or reservation was submitted for annexation to this report.

Under the Panel's constitutive instrument, the ballot is advisory and responsibility for the final determination rests with the Chair. Having considered the test evidence, the scope of the proposed pilot, the identified residual risks and the operational safeguards, Prof. Dr. Refrigério Sargaço granted conditional technical acceptance and authorised progression to the controlled pilot, subject to the scope limitations stated in this report.

07 Operational observation and operative effect

OPERATIONAL OBSERVATION

Anomaly SEC-014 concerns an intermittent timeout observed only during external API handshakes under extreme-load conditions. The event did not affect algorithmic integrity, data confidentiality or the correctness of the assessed decision outputs. It is therefore classified as a non-critical operational matter to be tracked through routine monitoring, capacity management and incident-recording procedures.

Accordingly, by decision of the Chair dated **30 April 2026**, the artefact may proceed to controlled pilot deployment within the approved scope. Any expansion beyond that scope requires separate authorisation and, where applicable, a further technical, security and compliance assessment.

OPERATIVE DECISION

CONDITIONAL TECHNICAL ACCEPTANCE

Final determination of the Panel Chair, dated 30 April 2026 and authorised exclusively for the controlled pilot described in this report. SEC-014 remains recorded as a non-critical operational observation.

Execution and acknowledgement

30 April 2026

GLOBAL EXECUTION DATE

The signatures below record: (i) issuance of the independent external report and the final determination by the President of Instituto Superior Técnico, acting as Chair of the Review Panel; (ii) acknowledgement of receipt by the commissioning entity; and (iii) governmental acknowledgement. The latter two signatures do not confer voting status and do not imply participation in the Panel's assessment or advisory ballot. This page forms an integral part of Independent Technical Assurance Report AMA-2026-0041-AI.

ISSUED AND DETERMINED



Prof. Dr. Refrigério Sargaço
President of Instituto Superior Técnico; Chair of the Independent IST Technical Review Panel

RECEIPT ACKNOWLEDGED



Dr. Eng. Armando Sacos
Chief Systems Architect, AMA; non-voting attendee

GOVERNMENTAL RATIFICATION



Eng. Só Craques
Minister for State Modernization; non-voting observer